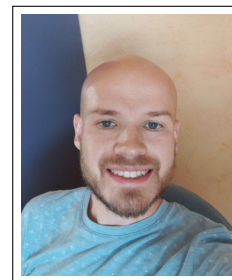


Thomas Jacumin

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Experiences

- September 2023–March 2027 **Postdoc in Applied Mathematics**, University Côte d’Azur, France, under the supervision of Pr. R. Masson
- March 2023–March 2025 **Postdoc in Applied Mathematics**, University of Lund, Sweden, under the supervision of Dr. A. Langer

Education

- 2019–2022 **PhD in Applied Mathematics**, University of Haute-Alsace, France, under the supervision of Pr. Z. Belhachmi
- 2017–2018 **Master in Modeling and Applied Mathematics**, University of Augsburg, Germany, with high honours
- 2016–2018 **Master in Applied Mathematics**, University of Rouen, France, with honours
- 2015–2016 **1st Year of Engineering School**, École nationale Supérieure en Informatique, Systèmes Avancés et Réseaux, Valence, France, validated
- 2013–2015 **DEUG of Mathematics “Prépa Écoles Ingénieurs”**, University of Avignon, France, with honours
- 2010–2013 **Baccalauréat Scientifique**, Lycée de l’Arc, Orange, France, with high honours

Publications

- Thomas Jacumin and Andreas Langer. “An Adaptive Finite Difference Method for Total Variation Minimization”. In: *Numerical Algorithms* (Mar. 19, 2025). ISSN: 1017-1398, 1572-9265. DOI: 10.1007/s11075-025-02044-6.
- Zakaria Belhachmi and Thomas Jacumin. “Adjoint Method in PDE-Based Image Compression”. In: *Asymptotic Analysis* (Oct. 2024), pp. 1–28. ISSN: 18758576, 09217134. DOI: 10.3233/ASY-241944.
- Zakaria Belhachmi and Thomas Jacumin. “Optimal Interpolation Data for PDE-Based Compression of Images with Noise”. en. In: *Communications in Nonlinear Science and Numerical Simulation* 109 (June 2022), p. 106278. ISSN: 1007-5704. DOI: 10.1016/j.cnsns.2022.106278.

Proceedings

- Thomas Jacumin et al. “A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media”. In: *ECMOR 2026*. ECMOR 2026. Porto, Portugal, European Association of Geoscientists & Engineers, 2026, pp. 1–20. DOI: 10.3997/2214-4609.202637078.

Preprints

- Zakaria Belhachmi and Thomas Jacumin. *Optimal Transport Model of Optical Flow Estimation: Constant and Varying Illumination Cases (accepted in Journal of Mathematical Study)*. Dec. 2024.
- Zakaria Belhachmi and Thomas Jacumin. *Iterative Approach to Image Compression with Noise: Optimizing Spatial and Tonal Data*. Sept. 29, 2022. DOI: 10.48550/arXiv.2209.14706. arXiv: 2209.14706.

Communications

- September 2026 **A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *ECMOR 2026*, Porto, Portugal
<https://eage.eventsair.com/ecmor-2026/>
- July 2026 **A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *CMWR 2026*, Bologna, Italy
<https://cmwrconference.org/>
- March 2026 **Adjoint Method in PDE-Based Image Compression**, *SMAI-MODE 2026*, Nice, France
<https://mode2026.sciencesconf.org/>
- March 2026 **Poster – A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *APPRISE Winter School*, Aussois, France
<https://apprise.sciencesconf.org>
- February 2026 **PDE-Based Images Compression**, *PhD Seminar*, University Côte d’Azur, France
- November 2024 **An Adaptive Finite Difference Method for Total Variation Minimization**, *Seminar*, University of Lund, Sweden
- January 2024 **Modèle Transport Optimal pour la Formulation du Flot Optique**, *ANR SOCOT*, University of Haute-Alsace, France
https://codimd.math.cnrs.fr/s/vKWD_JtAu
- July 2023 **Méthodes Adaptatives pour les Problèmes de Discontinuité**, *MAThEOR Days*, University of Strasbourg, France
<https://days.matheor.com/2023/>
- October 2022 **Mini-symposium – Inverse problems and shape optimization**, *Optimal interpolation data for PDE-based compression of images with noise*, *PICOF22* – University of Caen, France
<https://picof22.sciencesconf.org/>
- June 2022 **PhD Day – “Ma thèse en 180 secondes” format**, University of Haute-Alsace, France
- June 2022 **PhD Seminar**, *IRIMAS*, University of Haute-Alsace, France
- December 2021 **Poster**, *Calculus of Variations Colloquium*, University of Nancy, France
<https://indico.math.cnrs.fr/event/7000/>

Teaching

- January–April 2026 **Calculus 2**, *Exercice sessions intended for first-year undergraduate students in Applied Mathematics and Computer Science for the Social Sciences and Mathematics.*, University of Côte d’Azur, France
- March–November 2024 **Bachelor Thesis Supervision**, *Adaptive Finite Differences Method*, University of Lund, Sweden
- April–May 2022 **Master 1 Project Supervision**, *Image Compression by Image Inpainting*, University of Haute-Alsace, France

Work Groups

- November 2022 **Benamou-Brenier Formula**, *Optimal Transport*, University of Haute-Alsace, France
<https://juillet.perso.math.cnrs.fr/GTTO.html>
- March 2022 **C-transforms**, *Optimal Transport*, University of Haute-Alsace, France

Attended Schools

- March 2026 **Winter school APPRISE (Fundamentals of trAnsport Phenomena in PoRous medIa across ScalEs)**, Centre Paul-Langevin, France
- July 2023 **Summer school on Numerical Analysis of nonlinear PDEs**, University of Leipzig, Germany

Other Experiences

August 2022 **PhD Seminar Organization**, *Website Development*, IRIMAS, University of Haute-Alsace
June–July 2022 **Colloquium Organization**, *Website Development*, IRIMAS, University of Haute-Alsace
<http://www.lmia.uha.fr/colloque-lutz/>
June 2022 **Event Organization**, *Music Festival*, University of Haute-Alsace
September 2021 **AMIES Challenge**, *Trajectories Reconstruction*, Eurecam

Skills

French Native
English Good
Japanese Notions
IT Tools Python, SageMath, MATLAB, FreeFem++, Latex, C, C++, Java, VHDL, CUDA, Fortran, OpenFOAM

Referees

BELHACHMI Z., Professor,	IRIMAS, <i>University of Haute-Alsace</i> ,	<code>zakaria.belhachmi@uha.fr</code>
LANGER A., Assistant Professor,	LTH, <i>University of Lund</i> ,	<code>andreas.langer@math.lth.se</code>
MASSON R., Professor,	LJAD, <i>University Côte d’Azur</i> ,	<code>roland.masson@univ-cotedazur.fr</code>